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# ORDINARY DIFFERENTIAL Equations - MA417
## ASSIGNMENT 1
## Solution

Topic: Method of Solutions and Preliminaries

1.1 Let $y:[a,b]\to\mathbb{R}$ be a continuous function which satisfies the following relation $\frac{1}{2}y^2(t) \le \frac{y_0^2}{2} + \int_a^t \varphi(s)y(s)\,ds \ \forall t\in[a,b]$ where $a,b,y_0\in\mathbb{R}$ and $\varphi$ is a non-negative continuous function on $[a,b]$. Show that

$$|y(t)| \le |y_0| + \int_a^t \varphi(s)\,ds.$$

Sol$^n$: We have,

$$\frac{1}{2}y^2(t) \le \frac{y_0^2}{2} + \int_a^t \psi(s).y(s)\,ds. \quad (\text{where } y:[a,b]\to\mathbb{R}),$$

$\because$ $\psi(s) \geq 0$ then, either $\int_a^t \psi(s).y(s)\,ds$ can be positive or negative depending on $y(s)$.

Case I: When, $y(s)\geq 0$, on $[a,t]$

multiply 2 on both sides, we get

$$= y^2(t) \le y_0^2 + 2\int_a^t \psi(s).y(s)\,ds$$

Let $k_1(t) := y_0^2 + 2\int_a^t \psi(s).y(s)\,ds$ Then, $y^2(t)\le k_1(t)$

$y(t) \le |y(t)| \le \sqrt{k_1(t)}$, where $k_1(t)\geq 0$.

Note that,

$$k_1'(t) = 2\psi(t)y(t)$$
$$\le 2\psi(t)\sqrt{k_1(t)}$$

Now,

$$\frac{k_1'(t)}{2\sqrt{k_1(t)}} \le \psi(t) \Rightarrow \int_a^t d\left(\sqrt{k_1(s)}\right) \le \int_a^t \psi(s)\,ds$$

$$= \sqrt{k_1(t)} - \sqrt{k_1(a)} \le \int_a^t \psi(s)\,ds$$

$$= |y(t)| - |y_0| \le \int_a^t \psi(s)\,ds \quad (\because k_1(a) = y_0^2)$$

$$\Rightarrow |y(t)| \le |y_0| + \int_a^t \psi(s)\,ds.$$

Case II: Suppose $y(s)$ is negative.

we have,

$$\frac{y^2(t)}{2} \le \frac{y_0^2}{2} + \int_a^t \psi(s) \cdot y(s)\, ds.$$

multiplying 2 on both sides, we get

$$= y^2(t) \le y_0^2 + 2\int_a^t \psi(s) \cdot y(s)\, ds.$$

Now, since,

$$y(s) < 0$$

$$\Rightarrow \psi(s) \cdot y(s) < 0 \quad (\because \psi(s) \geq 0)$$

$$= 2\int_a^t \psi(s) \cdot y(s)\, ds < 0$$

$$\Rightarrow y^2(t) \le y_0^2 + 2\int_a^t \psi(s) \cdot y(s)\, ds < y_0^2$$

$$= |y(t)| < |y_0| \quad \text{--- (i)}$$

$$\because \int_a^t \psi(s)\, ds > 0 \Rightarrow |y_0| + \int_a^t \psi(s)\, ds > |y_0| \quad \text{--- (ii)}$$

using eq (i) & (ii), we get $|y(t)| < |y_0| + \int_a^t \psi(s)\, ds$. (Hence Proved)

1.2: Given,

$$x(t) \le x(\tau) + \int_\tau^t \phi(s) \cdot x(s)\, ds \;\forall\; t, \tau \in (a, b) \quad \text{--- (i)}$$

put $\tau = t_0 \in (a, b)$, we get

$$= x(t) \le x(t_0) + \int_{t_0}^t \phi(s) \cdot x(s)\, ds. \quad \text{--- (ii)}$$

Put $t = t_0$ & $\tau = t$ in eq(i), we get

$$= x(t_0) \le x(t) + \int_t^{t_0} \phi(s) \cdot x(s)\, ds$$

$$= x(t_0) \le x(t) - \int_{t_0}^t \phi(s) \cdot x(s)\, ds$$

$$\Rightarrow x(t) \geq x(t_0) + \int_{t_0}^t \phi(s) \cdot x(s)\, ds. \quad \text{--- (iii)}$$

using eq(i) & (ii), we get

$$= \quad x(t) = x(t_0) + \int_{t_0}^{t} \phi(s) \cdot x(s)\, ds.$$

Differentiation both side wrt. $t$ (with $t_0$ fixed), we get

$$= \quad x'(t) = \phi(t) \cdot x(t)$$

$$= \quad \frac{x'(t)}{x(t)} = \phi(t)$$

$$= \quad (\log(x(t)))' = \phi(t)$$

integrating both sides, we get

$$= \quad \int_{t_0}^{t} d(\log(x(s))) = \int_{t_0}^{t} \phi(s)\, ds$$

$$= \quad \log(x(t)) - \log(x(t_0)) = \int_{t_0}^{t} \phi(s)\, ds.$$

$$= \quad \log\left(\frac{x(t)}{x(t_0)}\right) = \int_{t_0}^{t} \phi(s)\, ds.$$

$$= \quad x(t) = x(t_0) \exp\left[\int_{t_0}^{t} \phi(s)\, ds\right]$$

for any fixed $t_0 \in (a,b)$.

<u>1.4</u> Given, $W(f,g)(t) = 3e^{4t}$ and $f(t) = e^{2t}$ $\forall t \in \mathbb{R}$.

Now, $W(f,g)(t) = \begin{vmatrix} f(t) & g(t) \\ f'(t) & g'(t) \end{vmatrix} = \begin{vmatrix} e^{2t} & g(t) \\ 2e^{2t} & g'(t) \end{vmatrix}$.

$$\Rightarrow \quad g'(t)\, e^{2t} - 2e^{2t} g(t) = 3e^{4t}$$

$$= \quad g'(t) - 2g(t) = 3e^{2t}.$$

$$= \quad g(t) \cdot e^{-2t} = \int 3\, dt \quad \Rightarrow \quad g(t) \cdot e^{-2t} = 3t + C$$

$$= \quad g(t) = (3t + C)\, e^{2t}$$

$$= \quad g(t) = (g(0) + 3t)\, e^{2t} \quad \forall t \in \mathbb{R}.$$

1.7. Given ODE,

$$u''(t) + a u'(t) + b u(t) = 0 \quad , t \in \mathbb{R}.$$

Let the characteristic polynomial be.

$$p(r) = r^2 + ar + b$$

characteristic roots:

$$r = \frac{-a \pm \sqrt{a^2 - 4b}}{2}$$

If $a^2 - 4b > 0$ then roots are real and distinct, say $r_1, r_2 \in \mathbb{R}$ $(r_1 \neq r_2)$.

The general solution

$$u(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$$

1. here for $u(t)$ to be bounded, both $r_1, r_2 \leq 0$

If $a^2 - 4b = 0$ then roots are real & equal.

then the general solution,

$$u(t) = (c_1 + c_2 t) e^{-a/2 t}$$

if $-\frac{a}{2} < 0 \Rightarrow a > 0$ then, $u(t)$ is bounded and non-trivial.

If $a^2 - 4b < 0$ then roots are of the form,

$r = \alpha \pm i\beta$, where $\alpha = -\frac{a}{2}$ and $\beta = \frac{\sqrt{4b - a^2}}{2}$

Then, general solution,

$$u(t) = e^{\alpha t}(c_1 \cos\beta t + c_2 \sin\beta t)$$

For $u(t)$ to be bounded and non-trival,

$$\alpha < 0 \Rightarrow a > 0$$

If $a = 0 \Rightarrow \alpha = 0$ and $u(t) = c_1 \cos\beta t + c_2 \sin\beta t$. which is bounded and non-trivial if $b > 0$

Thus, we conclude that for a bounded, non-trivial solution, we need $\boxed{a \geq 0}$.

Hence, the necessary and sufficient condition is $a \geq 0$ and $b \geq 0$ but not both $a=0$ and $b=0$.

($\because$ for both $a=0$ and $b=0$ the solution won't be defined as $u(t) \neq 0$)

1.8 Define for $t \in \mathbb{R}$, we get

$u_{i,j}(t) = t^j e^{\lambda_i t}$; $1 \leq i \leq m$; $0 \leq j \leq m_i$.

$v_{k,l}(t) = t^l e^{x_k t} \sin(y_k t)$; $1 \leq k \leq n$; $0 \leq l \leq n_k$.

$w_{l,k}(t) = t^l e^{x_k t} \cos(y_k t)$; $1 \leq k \leq n$; $0 \leq l \leq n_k$.

where $m, n, m_i, n_k$ are non-negative integers.

Now, to show that the set of functions $\{u_{i,j}(t), v_{k,l}(t), w_{l,k}(t)\}$ is linearly independent, we need to show linear combination.

$$\sum_{i=1}^{m} \sum_{j=0}^{m_i} a_{i,j} \cdot u_{i,j}(t) + \sum_{k=1}^{n} \sum_{l=0}^{n_k} b_{k,l} v_{k,l}(t) + \sum_{k=1}^{n} \sum_{l=0}^{n_k} c_{l,k} w_{l,k}(t) = 0 \quad \ldots (*)$$

has all coefficients $a_{i,j}, b_{k,l}, c_{l,k}$ must necessarily be zero.

Now, the only way sum $(*)$ can be identically zero is if each group of functions (real exponential times polynomial; oscillatory trigonometric functions) independently sums to zero i.e.

$$\sum a_{i,j} t^{i} e^{\lambda_i t} = 0 \; ; \; \sum b_{k,l} t^{l} e^{x_k t} \sin(y_k t) = 0 \; ;$$

$$\sum c_{l,k} t^{l} e^{x_k t} \cos(y_k t) = 0 .$$

Thus, the only solution is the trivial one.
i.e. all coefficient zero.

i.e. $a_{i,j} = b_{k,l} = c_{l,k} = 0$.

Hence, the set $\{ u_{i,j}(t), v_{k,l}(t), w_{l,k}(t) \}$ is linearly independent.